

Name: _____

SOLUTIONS

Date: _____

Year 8 Mathematics

Test 2, 2015

Topics – Length, Perimeter and Area

60 = _____ %
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Total Time: 60 minutes

SECTION 1: CALCULATOR FREE

For each question, subtract $\frac{1}{2}$ mark for no units, and $\frac{1}{2}$ for rounding

Time: 25 minutes	Total Marks: 26	Equipment Allowed: Nil
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1. [2 marks: $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{2}$]

For each of the following questions, shade the bubble below the answer you think is most correct.

a) How many metres are there in a kilometre?

10

☐

100

☐

1000

☒

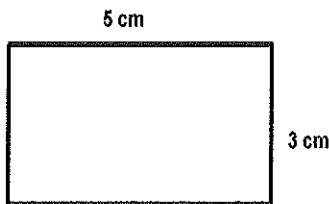
10 000

☐

100 000

☐ $\frac{1}{2}$.

Questions b) and c) relate to the diagram below



b) The perimeter of this rectangle above is:

15 cm

☐

8 cm

☐

30 cm

☐

12 cm

☐

16 cm

☒ $\frac{1}{2}$.

c) The area of the rectangle above is:

15 cm²☒16 cm²☐20 cm²☐8 cm²☐35 cm²☐ $\frac{1}{2}$.

d) If the length of a parallelogram's base was 10cm and its height was 4 cm what would its area be?

40 cm

☐

14 m

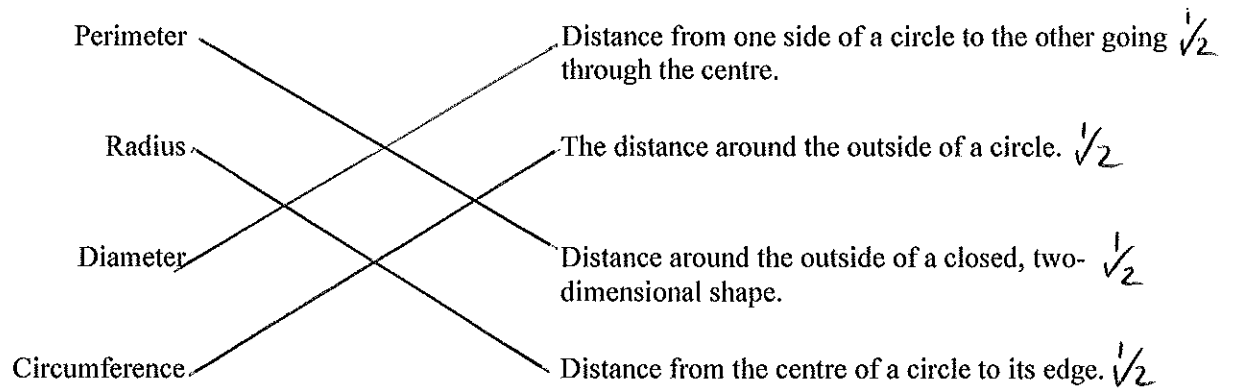
☐

28 cm

☐40 cm²☒28 cm²☐ $\frac{1}{2}$.

2. [2 marks: $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{2}$]

Match these words with the most accurate meaning.



3. [6 marks: 1, 1, 2, 2]

Complete the following conversions:

a) $52 \text{ cm} = \underline{0.52} \text{ m} \checkmark$

b) $1.65 \text{ km} = \underline{1650} \text{ m} \checkmark$

c) $980 \text{ mm} = \underline{0.98} \text{ m} \checkmark\checkmark$

d) $0.38 \text{ km} = \underline{38000} \text{ cm} \checkmark\checkmark$

4. [6 marks: 1, 1, 2, 2]

Complete the following conversions:

a) $95 \text{ mm}^2 = \underline{0.95} \text{ cm}^2 \checkmark$

b) $2.45 \text{ m}^2 = \underline{24500} \text{ cm}^2 \checkmark$

c) $75\,789 \text{ mm}^2 = \underline{0.075789} \text{ m}^2 \checkmark\checkmark$

d) $0.028 \text{ km}^2 = \underline{2800000000} \text{ cm}^2 \checkmark\checkmark$

5. [2 marks]

If a rectangle has an area of 32 cm^2 , give 2 possible sets of dimensions it could have.

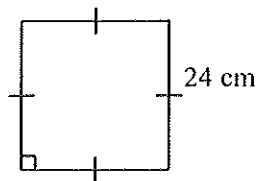
1×32
 2×16
 4×8

$\checkmark$ per correct set of dimensions.

6. [8 marks: 1, 2, 2, 3]

Calculate the perimeter of the following shapes:

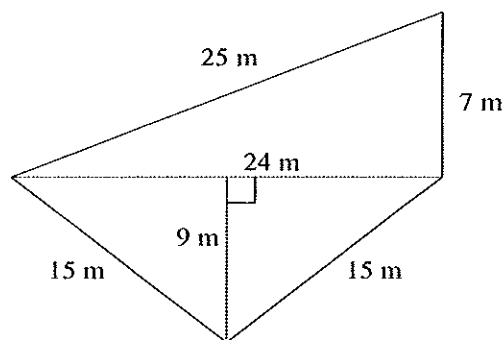
a)



$$P = 24 \times 4$$

$$= 96 \text{ cm} \checkmark$$

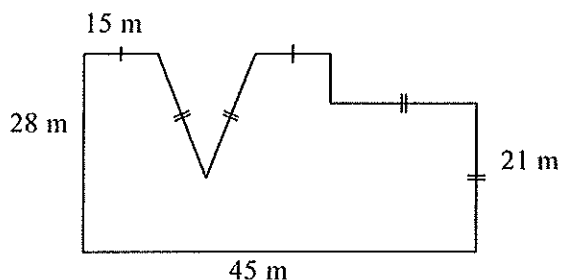
b)



$$P = 15 + 25 + 15 + 7 \checkmark$$

$$= 62 \text{ m} \checkmark$$

c)

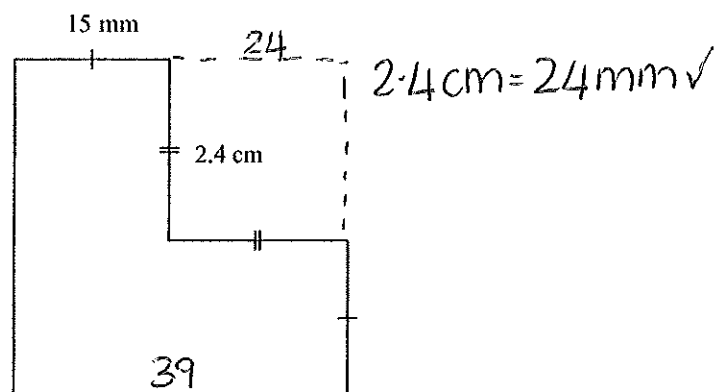


$$P = 15 \times 2 + 21 \times 4 + 28 + 45 \checkmark$$

$$= 30 + 84 + 28 + 45$$

$$= 187 \text{ m} \checkmark$$

d)



$$P = 39 \times 4 \checkmark$$

$$= 156 \text{ mm} \checkmark$$

$$= 15.6 \text{ cm}$$

or

$$P = 15 \times 2 + 24 \times 2 + 39 + 39 \checkmark$$

$$= 156 \text{ mm} \checkmark$$

$$= 15.6 \text{ cm}$$

~ END OF TEST SECTION 1 ~

Name: _____

Date: _____

SECTION 2: CALCULATOR ASSUMED

Time: 35 minutes	Total Marks: 34	Equipment Allowed: <i>½ page notes (A4 one side), Scientific calculator</i>
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7. [10 marks: 1, 2, 1, 2, 2, 2]

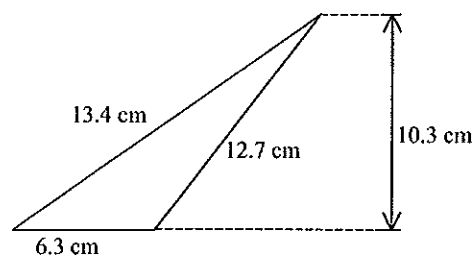
Determine the area of the following:

a) Square with side lengths of 7.5 m

$$A = 7.5 \times 7.5$$

$$= 56.25 \text{ m}^2 \checkmark$$

b)



$$A = \frac{1}{2} \times 6.3 \times 10.3 \checkmark$$

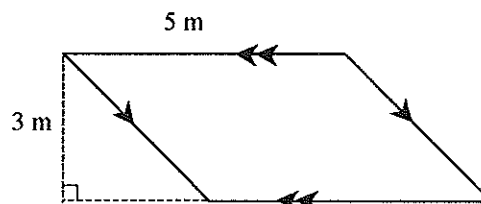
$$= 32.445 \text{ cm}^2 \checkmark$$

c) Rectangle with dimensions 3.2 cm by 2.1 cm.

$$A = 3.2 \times 2.1$$

$$= 6.72 \text{ cm}^2 \checkmark$$

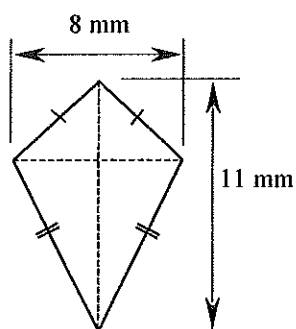
d)



$$A = 5 \times 3 \checkmark$$

$$= 15 \text{ m}^2 \checkmark$$

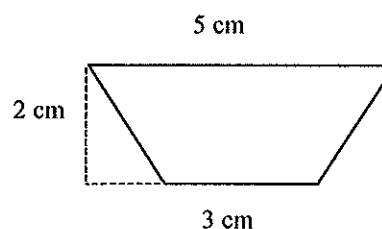
e)



$$A = \frac{1}{2} \times 8 \times 11 \checkmark$$

$$= 44 \text{ mm}^2 \checkmark$$

f)

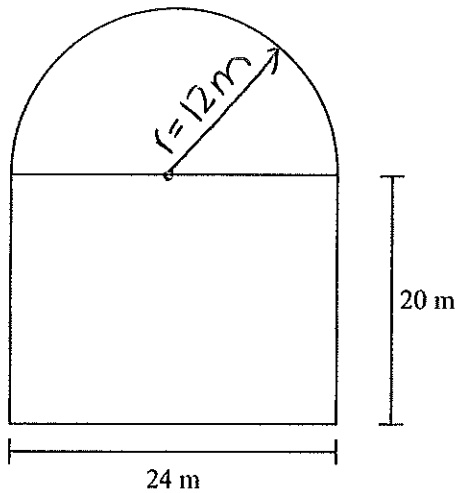


$$A = \frac{1}{2} (3 + 5) \times 2 \checkmark$$

$$= 8 \text{ cm}^2 \checkmark$$

8. [3 marks]

Determine the area of the arch window below. Give your answer correct to two decimal places.



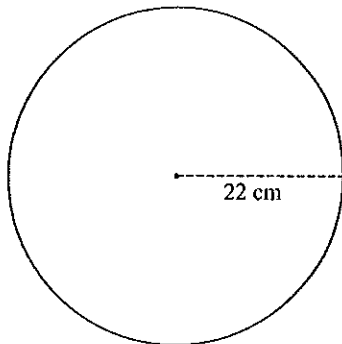
$$A = \frac{1}{2} \times \pi \times 12^2 + 24 \times 20 \checkmark \checkmark$$

$$= 706.19 \text{ m}^2 \checkmark$$

9. [4 marks: 2, 2]

Determine the perimeter of the following. Round your answer correct to two decimal places.

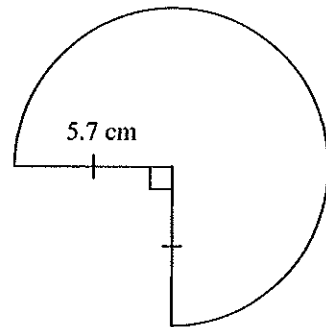
a)



$$P = 2 \times \pi \times 22 \checkmark$$

$$= 138.23 \text{ cm (2dp)} \checkmark$$

b)



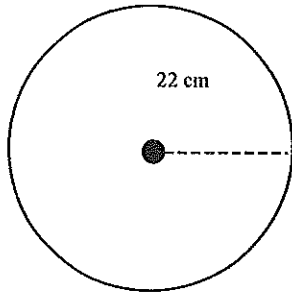
$$P = \frac{3}{4} \times 2 \times \pi \times 5.7 + 2 \times 5.7 \checkmark$$

$$= 38.26 \text{ cm (2dp)} \checkmark$$

10. [4 marks: 2, 2]

Determine the area of the following, correct to one decimal place.

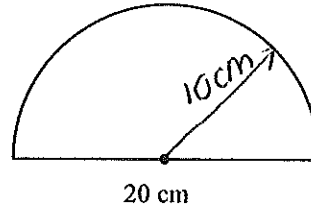
a)



$$A = \pi \times 22^2 \quad \checkmark$$

$$= 1520.5 \text{ cm}^2 \text{ (1dp)} \quad \checkmark$$

b)



$$A = \frac{1}{2} \times \pi \times 10^2 \quad \checkmark$$

$$= 157.1 \text{ cm}^2 \text{ (1dp)} \quad \checkmark$$

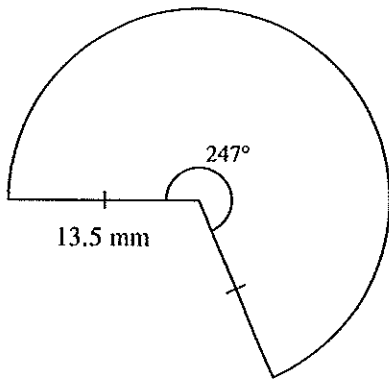
11. [4 marks: 2, 2]

Determine the perimeter and area of the following sector, correct to one decimal place.

a) Perimeter:

$$= \frac{247}{360} \times 2 \times \pi \times 13.5 + 2 \times 13.5 \quad \checkmark$$

$$= 85.2 \text{ mm (1dp)} \quad \checkmark$$



b) Area:

$$= \frac{247}{360} \times \pi \times 13.5^2 \quad \checkmark$$

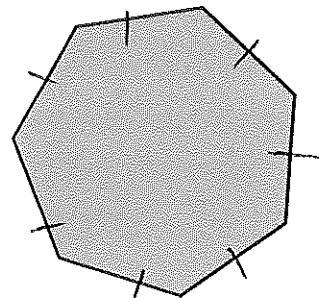
$$= 392.8 \text{ mm}^2 \text{ (1dp)} \quad \checkmark$$

12. [2 marks]

A regular polygon shown has a perimeter of 112 m.
How long is each side?

$$\text{side} = 112 \div 7 \quad \checkmark$$

$$= 16 \text{ m} \quad \checkmark$$



13. [2 marks]

A circle has a circumference of 18 m.

What is its radius, correct to three decimal places?

$$C = 2 \times \pi \times r$$

$$18 = 2 \times \pi \times r$$

$$r = 18 \div 2 \div \pi \quad \checkmark$$

$$r = 2.865 \text{ m (3dp)} \quad \checkmark$$

14. [3 marks]

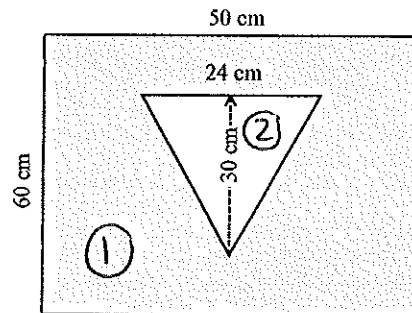
A triangle has been cut out of a rectangular piece of card. What area of card remains?

$$A \text{ ①} = 60 \times 50$$

$$= 3000 \text{ cm}^2 \quad \checkmark$$

$$A \text{ ②} = \frac{1}{2} \times 24 \times 30$$

$$= 360 \text{ cm}^2 \quad \checkmark$$



$$\therefore \text{shaded} = 3000 - 360 = 2640 \text{ cm}^2 \quad \checkmark$$

15. [2 marks]

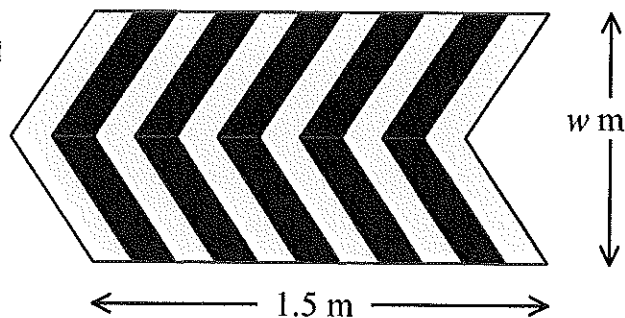
A warning sign is in the shape of an arrow made up of two parallelograms, as shown.

The length of one edge of the sign is 1.5 m and an area of 1.8 m².

What is the width of the sign (marked w)?

$$\text{height} = 1.8 \div 1.5$$

$$= 1.2 \text{ m} \quad \checkmark$$



$$\therefore \text{width} = 2 \times 1.2 = 2.4 \text{ m} \quad \checkmark$$

~ END OF TEST SECTION 2 ~